1 Synthesis: Motion and Patterns

Central question. How can the different mathematical languages of this course describe one evolving system?

1.1 One motion, several descriptions

Imagine a point traveling counterclockwise around a circle of radius R with angular speed ω. Its angle at time t is $\theta(t) = \theta_0 + \omega t$. In parametric form,

$$x(t) = R\cos(\theta_0 + \omega t), \qquad y(t) = R\sin(\theta_0 + \omega t).$$

Its polar form is $(r, \theta) = (R, \theta_0 + \omega t)$, and its complex form is

$$\boxed{z(t) = R\big(\cos(\theta_0 + \omega t) + i\sin(\theta_0 + \omega t)\big)}.$$

These are not competing models: they encode the same physical position in different coordinate languages.

1.2 Velocity, vectors, and periodicity

Over a short time interval Δt, the displacement vector is

$$\Delta\mathbf{r} = \langle x(t + \Delta t) - x(t), y(t + \Delta t) - y(t)\rangle.$$

For constant angular speed the point completes a revolution after $T = 2\pi/|\omega|$ when $\omega \neq 0$. Its horizontal and vertical coordinates repeat with that period. The direction of motion is tangent to the circle and perpendicular to the radius: this is the geometric heart of circular motion.

1.3 Sampling creates a sequence

Suppose we record the position every Δt seconds. The resulting sequence of complex positions is

$$z_n = R\big(\cos(\theta_0 + n\delta) + i\sin(\theta_0 + n\delta)\big), \qquad \delta = \omega\Delta t.$$

Writing $u = \cos\delta + i\sin\delta$ gives

$$\boxed{z_{n+1} = uz_n, \qquad z_n = u^n z_0.}$$

De Moivre's theorem then explains every sample as a repeated rotation. If $\delta = 2\pi/m$ for a positive integer m, the sampled orbit closes after m steps (provided this is the smallest such positive step count). If $\delta/(2\pi)$ is irrational, no two different samples have exactly the same position.

1.4 Scaling and rotation together

Now consider the recurrence $z_{n+1} = qz_n$ where

$$q = \rho(\cos\delta + i\sin\delta).$$

Its explicit solution is $z_n = q^n z_0$. Consequently,

$$|z_n| = \rho^n|z_0|, \qquad \arg z_n \equiv \arg z_0 + n\delta \pmod{2\pi}$$

for nonzero terms. If $0 < \rho < 1$, the points spiral toward the origin; if $\rho > 1$, they spiral outward; and if $\rho = 1$, they remain on a circle. Thus a geometric sequence and a rotation combine to produce a spiral.

1.5 Sums represent accumulated effects

If a system moves by step vectors $\mathbf{v}_0, \mathbf{v}_1, \ldots$, its total displacement after N steps is the vector sum $\sum_{n=0}^{N-1} \mathbf{v}_n$. For a complex-valued geometric pattern $\Delta z_n = aq^n$ with $q \neq 1$,

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q}.$$

Notice that the location sequence z_n and the step sequence Δz_n are different objects. Careful modeling starts by distinguishing what is being measured.

1.6 Different curves arise from different constraints

Circular and elliptical motion can be expressed parametrically. A projectile follows a parabola when horizontal velocity is constant and vertical acceleration is constant. Polar formulas express rays and radial patterns directly. The useful representation depends on whether we need a shape, a position, a direction, or a rule for change.

1.7 A complete example

A sensor begins at $z_0 = 2$ in the complex plane. Every second its controller multiplies the position by $q = \frac{1}{2}(\cos 60^\circ + i \sin 60^\circ)$. After n steps,

$$z_n = 2\left(\frac{1}{2}\right)^n (\cos(60^\circ n) + i \sin(60^\circ n)).$$

After three steps the modulus is $1/4$, and the direction is 180°. Hence $z_3 = -1/4$. The modulus obeys a geometric sequence, the direction obeys an arithmetic progression of angles, and the position follows a discrete inward spiral.

1.8 Exercises

1. Write Cartesian, polar, and complex formulas for a circle of radius 5 rotating once every 20 seconds, starting on the positive horizontal axis.

2. Sample that motion every five seconds. Give its first five complex positions and identify its repetition period in samples.

3. For $z_{n+1} = iz_n$ and $z_0 = 1 + i$, calculate $z_1, \ldots, z_4$ and describe the geometry.

4. For $z_{n+1} = (1/2)iz_n$, $z_0 = 4$, find z_4 and explain the change in magnitude and argument.

5. A point travels along $x = 3\cos t$, $y = 2\sin t$. Eliminate the parameter and describe one complete traversal.

6. Explain why a constant-speed circular trajectory is periodic but the spiral $z_n = (0.8e^{i\pi/3})^n z_0$ is not periodic when $z_0 \neq 0$.

7. Compute the finite sum $1 + i + i^2 + i^3$ and interpret its geometric cancellation.

8. Describe a real-world process that could be analyzed with at least three of the course's mathematical representations.